

Strong-lens parameter inference: EM-only, GW-only, and EM+GW

A cross-method comparison on a common mock

July 18, 2026

1 System and simulation

All three cases fit the *same* mock lens (`shared/system.config.py`, seed 87651): an EPL+SHEAR deflector ($\theta_E = 1.2''$, $e_1 = 0$, $e_2 = 0.1$, $\gamma = 2$, shear $\gamma_1 = 0.1$, $\gamma_2 = 0$) with a Sérsic source at $(0.2, -0.05)''$ and a Sérsic lens light, imaged on a 40×40 grid at $0.1''/\text{px}$, Gaussian PSF (FWHM $0.067''$), background rms 10^{-2} , $t_{\text{exp}} = 2200 \text{ s}$. Peak SNR ≈ 290 . The GW sector adds four lensed images with observables = relative time delays (3) + effective luminosity distances (4); truth $T_\star = 1.479 \times 10^7 \text{ s}$, $d_L = 11214 \text{ Mpc}$.

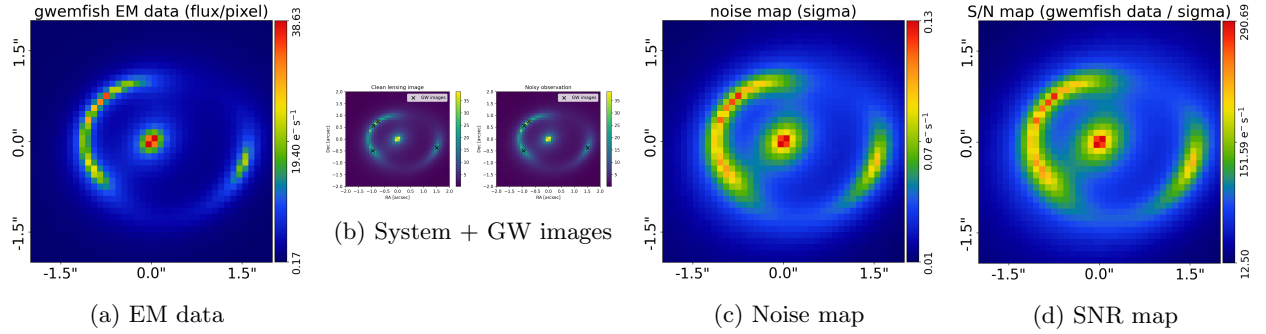


Figure 1: Simulated observation. (a) noisy EM image; (b) clean+noisy image with the 4 GW image positions ($\times$); (c) per-pixel noise; (d) per-pixel SNR (peak ≈ 290).

Measurement-error regimes (GW sector). Two settings are compared for the GW cases: *large-error* ($\sigma_{t_d} = 5\%$, $\sigma_{d_L^{\text{eff}}} = 300\%$) and *precise* ($\sigma_{t_d} = 0.1\%$, $\sigma_{d_L^{\text{eff}}} = 5\%$). The simulated observables are identical; only the assumed uncertainties (posterior widths) change. EM-only is unaffected.

2 Case 1 — EM-only

Free (11): $\theta_E, e_1, e_2, \gamma, \gamma_1, \gamma_2$ and source A, R_S, n, e_1, e_2 . **Fixed to truth:** lens + source centres, shear origin, full lens-light profile, background noise. **Methods:** gwemfish deriv-approx (informed NUTS on a Taylor model) and fisher (Taylor–Gaussian at truth); PyAutoLens Nautilus; lenstronomy Nautilus. Priors: gwemfish/PAL use their native broad priors, lenstronomy uses truth $\pm 10\sigma_{\text{Fisher}}$ boxes; at SNR ~ 290 the fit is data-dominated (verified by overlap).

Result. All four frameworks agree on every parameter to $\lesssim 0.2\sigma$ in the means, with widths matching to a few percent (Table 1). The only outlier is source amplitude/radius ($\sim 1\sigma$, $\sim 0.2\%$): the irreducible Sérsic b_n -convention difference (PAL Ciotti–Bertin vs. HCL/lenstronomy). Common ~ 1 – 2σ pulls ($\theta_E - 1.2$, $n + 2.1$) are the shared noise realization, not bias — the truth-centred Fisher posterior has identical widths.

Table 1: Case 1, representative parameters (mean $\pm$ std; pull = (mean–truth)/std).

param (truth)	gwemfish deriv	gwemfish fisher	PyAutoLens	lenstronomy
θ_E (1.2)	$1.1998 \pm 1.6\text{e-}4$ (−1.2)	$1.2000 \pm 1.6\text{e-}4$ (0.0)	$1.1998 \pm 1.6\text{e-}4$ (−1.2)	$1.1998 \pm 1.6\text{e-}4$ (−1.2)
γ (2.0)	$2.0005 \pm 5.2\text{e-}4$ (+0.9)	$2.0000 \pm 5.4\text{e-}4$ (0.0)	$2.0004 \pm 5.4\text{e-}4$ (+0.8)	$2.0004 \pm 5.4\text{e-}4$ (+0.8)
e_2 (0.1)	$0.1000 \pm 3.8\text{e-}4$ (−0.1)	$0.1000 \pm 3.7\text{e-}4$ (0.0)	$0.1000 \pm 3.7\text{e-}4$ (−0.1)	$0.1000 \pm 3.8\text{e-}4$ (0.0)
n_{src} (1.5)	$1.5044 \pm 2.0\text{e-}3$ (+2.2)	$1.5000 \pm 2.0\text{e-}3$ (0.0)	$1.5039 \pm 2.0\text{e-}3$ (+2.0)	$1.5041 \pm 2.0\text{e-}3$ (+2.1)

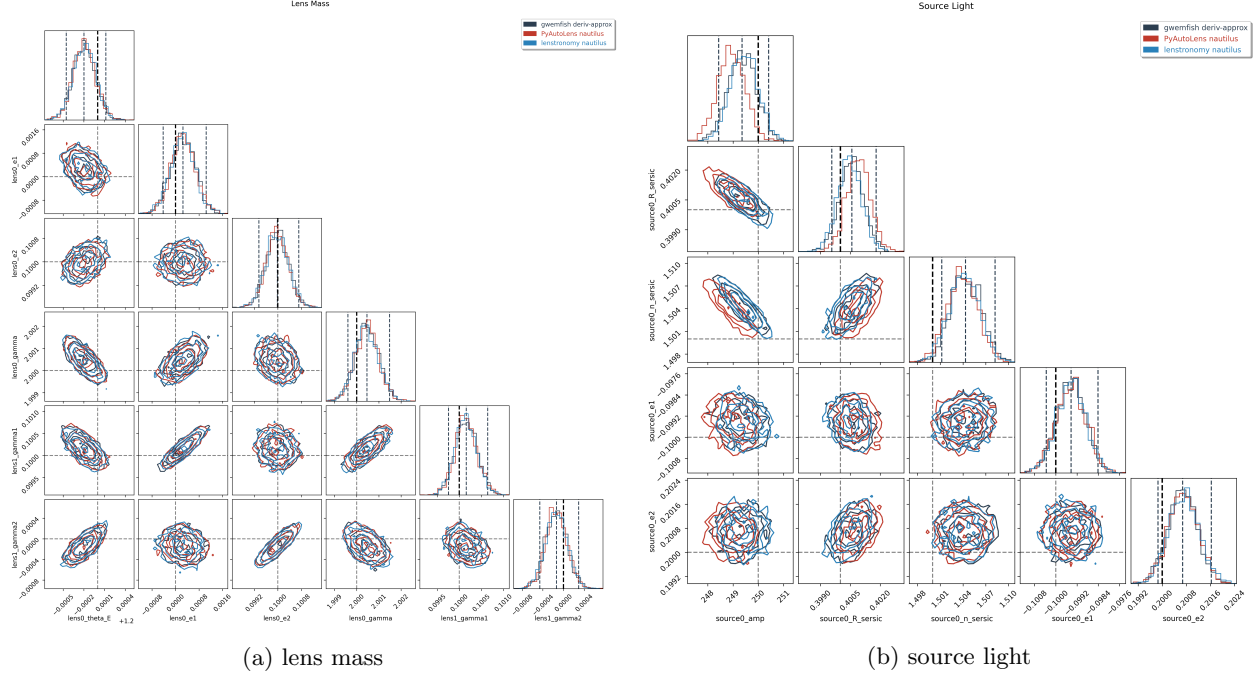


Figure 2: Case 1 EM-only: four-way comparison corners. Contours overlap to $\lesssim 0.2\sigma$.

3 Case 2 — GW-only

Free (4): $e_2, \gamma, y_0^{\text{gw}}, y_1^{\text{gw}}$ (source position). **Fixed to truth:** θ_E, e_1 , lens centre, all shear, $T_\star, d_L$. Constraint = time delays (d_L^{eff} nearly uninformative in large-error). **Methods:** fisher-source, deriv-approx-source, nautilus-source (helens solver), and nautilus with a lenstronomy solver (same GW likelihood, different lens-equation solver — a cross-check). **Priors (Uniform):** NUTS boxes $e_2 \in (-0.5, 0.5)$, $\gamma \in (1.5, 2.5)$, $y_0 \in (0.1, 0.3)$, $y_1 \in (-0.13, 0.03)$; nautilus uses truth $\pm 3\sigma_{\text{Fisher}}$ clipped to these (tight in precise, NUTS-dominated in large-error).

Result. Every method is consistent with truth (all $|\text{pull}| \leq 0.7$). Precise data tighten the posteriors $\sim 5\text{--}40\times$ (Table 2) and *resolve the y_1^{gw} bimodality* seen in large-error (Fig. 3). Fisher-source flips from a poor, $\sim 10\times$ -too-wide approximation (near-singular FM, cond 1.9×10^9) to an excellent one. The two solvers agree to $\sim 10^{-8}$ nats except on a $\sim 5\text{--}10\%$ caustic-boundary sliver each truncates differently ($\leq 0.4\sigma$ mean shifts).

Table 2: Case 2 GW-only, posterior std (nautilus + lenstronomy solver); all means at truth.

param	large-error std	precise std	tighter
e_2	0.029	0.0032	$\sim 9\times$
γ	0.216	0.047	$\sim 5\times$
y_0^{gw}	0.0097	0.0011	$\sim 9\times$
y_1^{gw}	0.0186	0.0005	$\sim 40\times$

4 Case 3 — EM+GW

Free (27): full joint EM pixel model + GW sector ($T_\star, d_L, y_0^{\text{gw}}, y_1^{\text{gw}}$). **Fixed to truth:** shear origin only. **Methods:** fisher-source, deriv-approx-source, nautilus-source (helens). **Priors:** lens-light centroid $\mathcal{N}(0, 0.05)$, $y_0/y_1 \sim \text{Uniform truth} \pm 0.05$, rest free per parameter layout; nautilus boxes truth $\pm 5\sigma_{\text{Fisher}}$. (nautilus ties the GW source to the EM source centre, so its y_0/y_1 column carries the EM astrometric constraint.)

Result. The EM likelihood already pins the mass and source-position parameters, so precise GW data

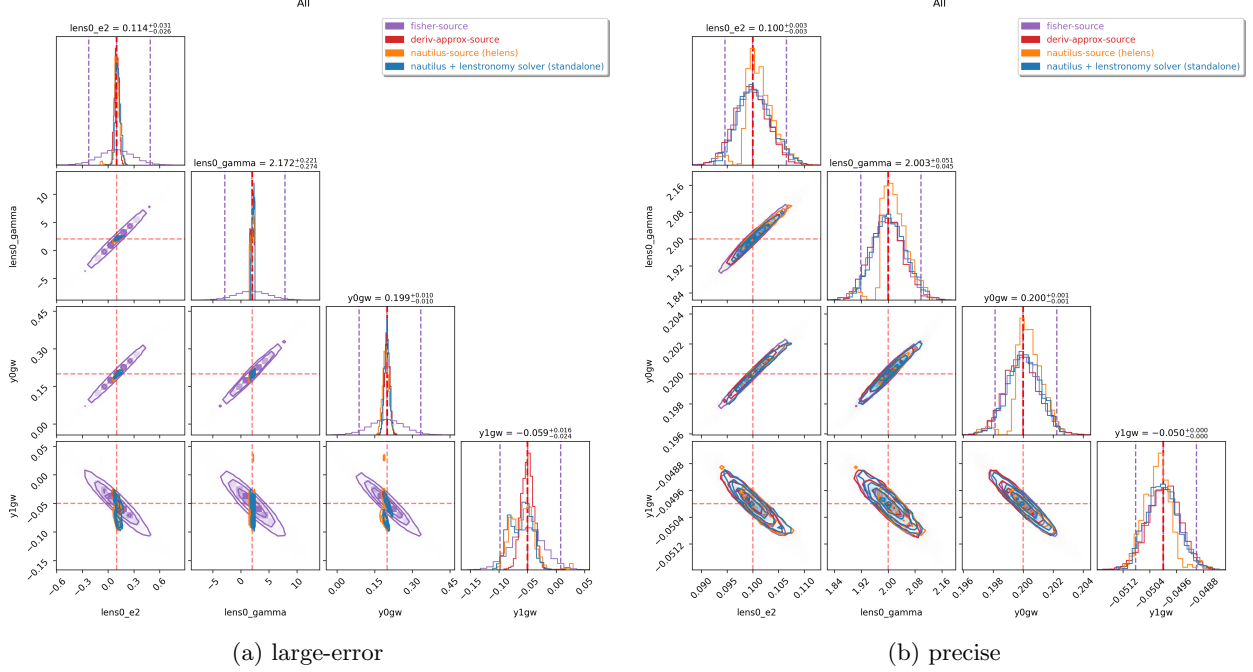


Figure 3: Case 2 GW-only comparison corners. Precise data collapse the widths and remove the secondary y_1^{gw} mode near -0.08 visible at large error.

change them negligibly; what they transform is the *GW-only sector* (Table 3): d_L goes from effectively unconstrained (± 11800 , $\sim 70\%$) to ± 288 Mpc ($\sim 2.6\%$), and $T_\star$ tightens $\sim 11\times$. fisher and nautilus recover truth (nautilus $|\text{pull}| \leq 1.2$; large pulls are std-amplification, not bias — absolute accuracy is sub-mas). *Caveat*: deriv-approx (banana surrogate) mixes poorly on the sharp precise 27-D posterior (worst $\hat{R} \approx 1.9$); treat its contour as indicative — fisher and nautilus are the reliable methods here.

Table 3: Case 3 EM+GW, nautilus-source std: where precise GW data help.

param	large-error std	precise std	tighter
d_L (Mpc)	11800 (unconstr.)	288	$\sim 40\times$
$T_\star$ (s)	4.9×10^5	4.6×10^4	$\sim 11\times$
γ	0.0030	0.0028	$\sim 1\times$
y_1^{gw}	0.00020	0.00019	$\sim 1\times$

5 Conclusion

On a common mock, independent codes and inference methods agree: (i) **EM-only** — gwemfish, PyAutoLens and lenstronomy recover truth to $\lesssim 0.2\sigma$, the only residual being the known Sérsic b_n -convention offset; (ii) **GW-only** — all methods are truth-consistent; precise timing shrinks the posterior $\sim 5\text{--}40\times$ and resolves the y_1^{gw} bimodality, and Fisher becomes an accurate surrogate; (iii) **EM+GW** — EM fixes the lens/source, so precise GW data act almost entirely on the GW-only sector, turning d_L from unconstrained into a $\sim 2.6\%$ measurement. Cross-checks hold throughout (independent solvers agree to $\sim 10^{-8}$ nats; vectorized vs. scalar likelihoods to $\sim 10^{-11}$). The one method-level caveat is deriv-approx-source, whose banana-surrogate NUTS mixes poorly once the EM+GW posterior becomes very sharp — fisher-source and nautilus-source remain reliable across all regimes.

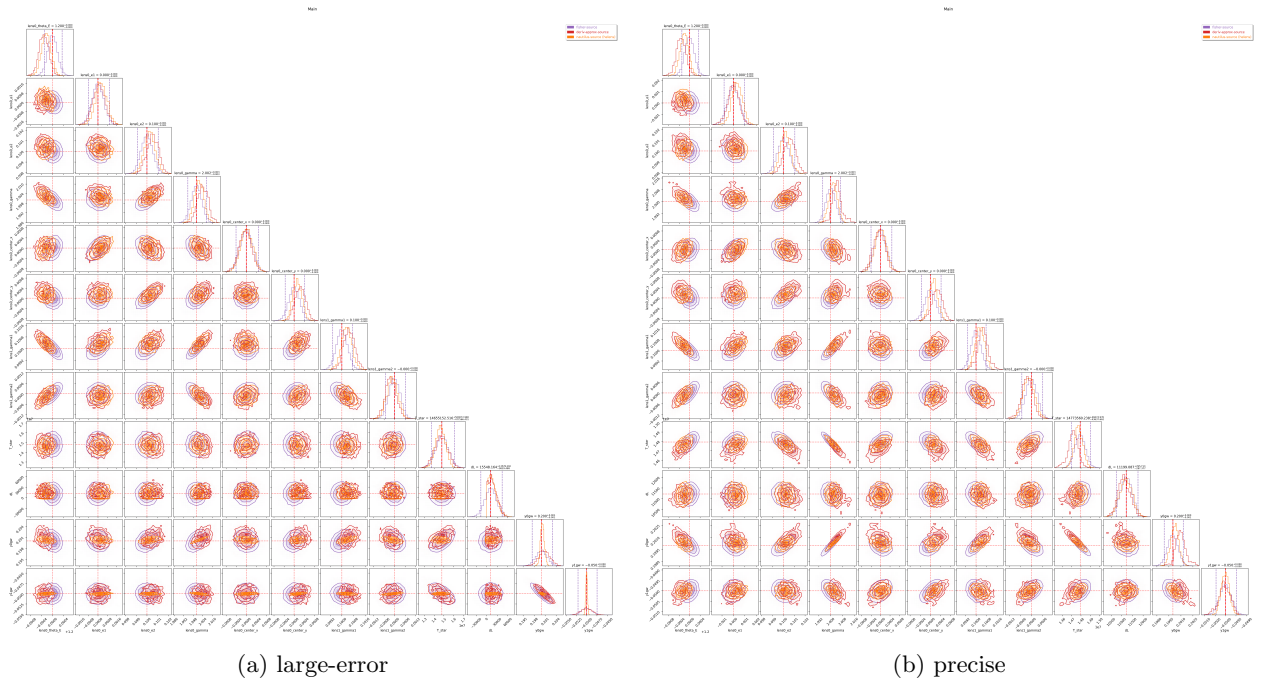


Figure 4: Case 3 EM+GW comparison corners (main parameters). Mass/source params are EM-dominated and stable; the visible tightening is in the GW sector ($d_L, T_\star$).